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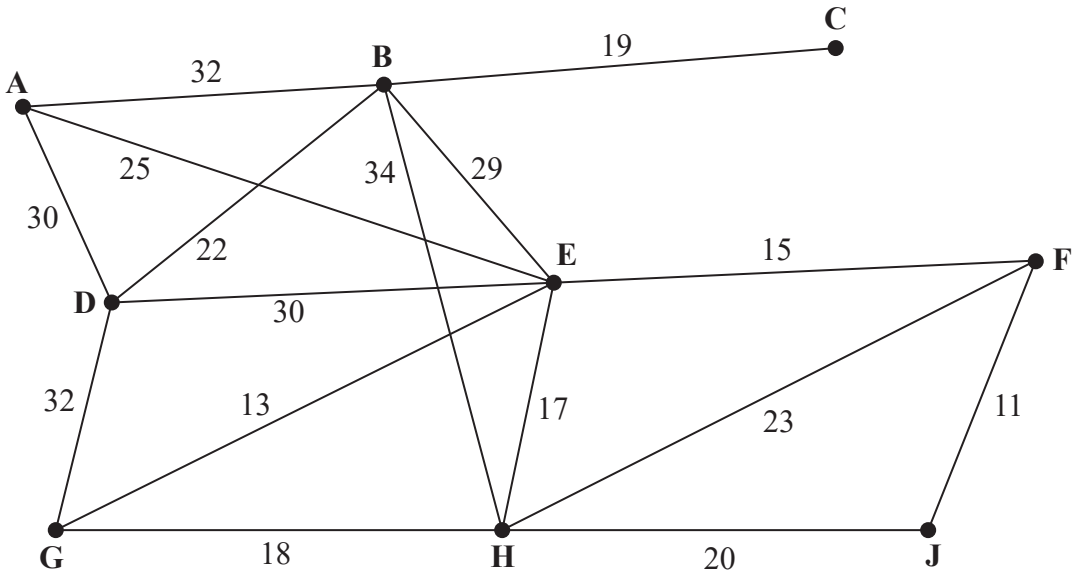


Figure 1

(a) Define the terms

(i) tree,

(ii) minimum spanning tree.

(3)

(b) Use Kruskal's algorithm to find the minimum spanning tree for the network shown in Figure 1. You must clearly show the order in which you consider the edges. For each edge, state whether or not you are including it in the minimum spanning tree.

(3)

(c) Draw the minimum spanning tree using the vertices given in Diagram 1 in the answer book and state the weight of the minimum spanning tree.

(2)

(Total 8 marks)

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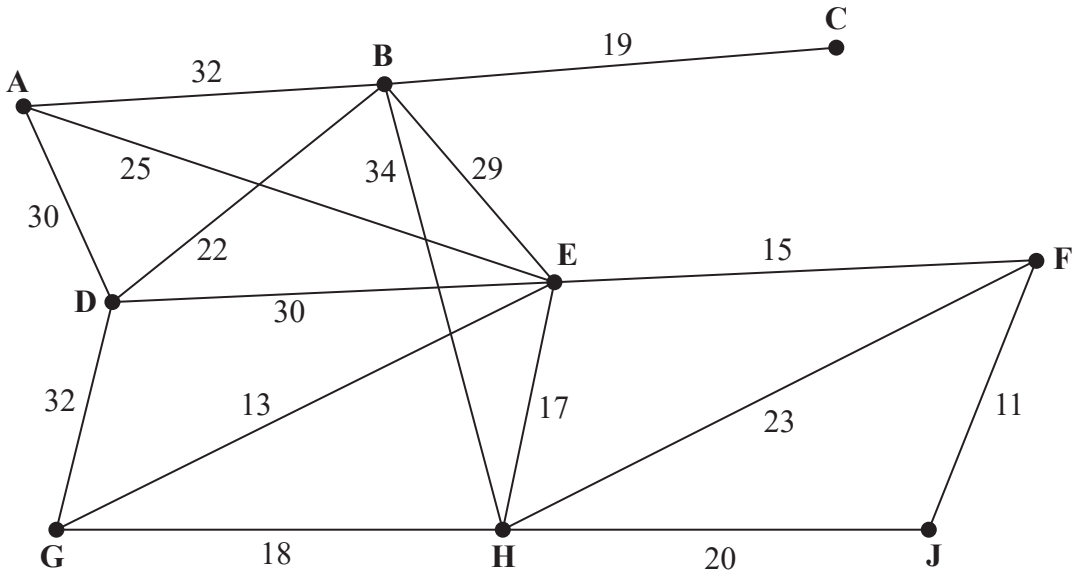


Figure 1

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Question 2 continued

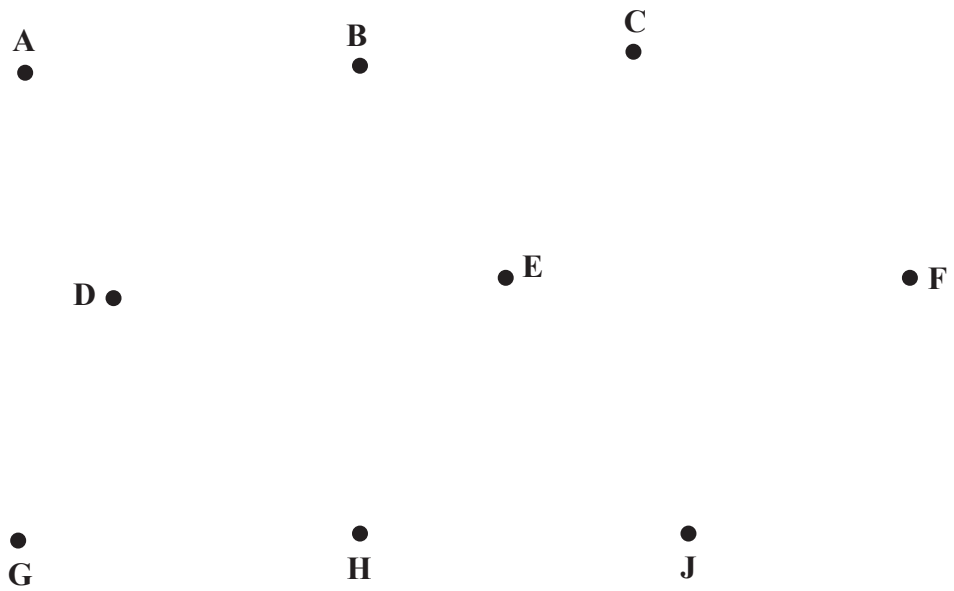


Diagram 1

Weight of the minimum spanning tree _____

Q2

(Total 8 marks)

2.

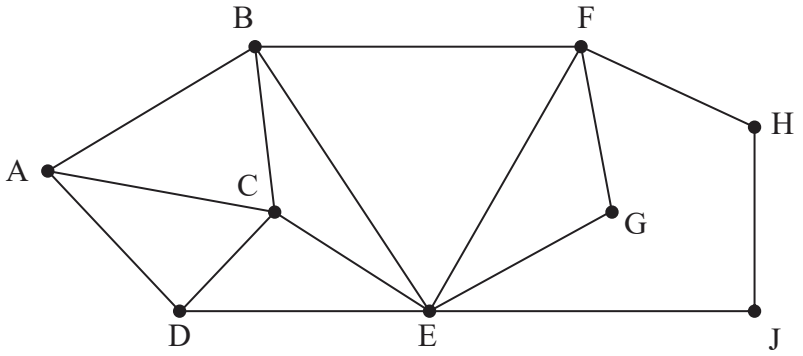


Figure 1

Figure 1 shows a graph, T.

(a) Write down an example of a path from A to J on T.

(1)

(b) State, with a reason, whether A – B – C – D – E – G – F – H – J is an example of a tour on T.

(1)

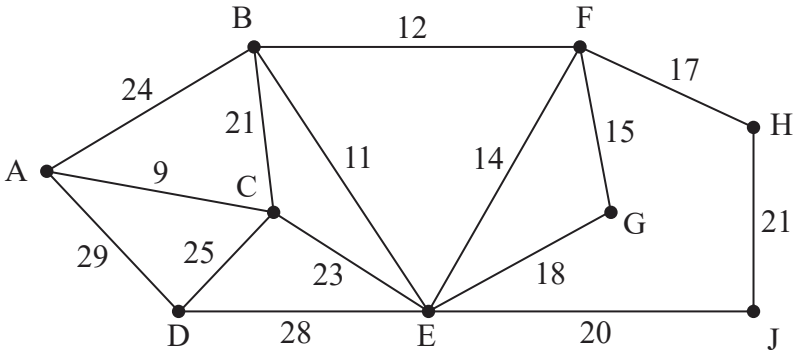


Figure 2

The numbers on the 15 arcs in Figure 2 represent the distances, in km, between nine vertices, A, B, C, D, E, F, G, H and J, in a network.

(c) Use Kruskal's algorithm to find the minimum spanning tree for the network. You should list the arcs in the order in which you consider them. In each case, state whether or not you are adding the arc to the minimum spanning tree.

(3)

(d) Draw the minimum spanning tree using the vertices given in Diagram 1 in the answer book.

(1)

(e) State the weight of the minimum spanning tree.

(1)

(Total 7 marks)

2.

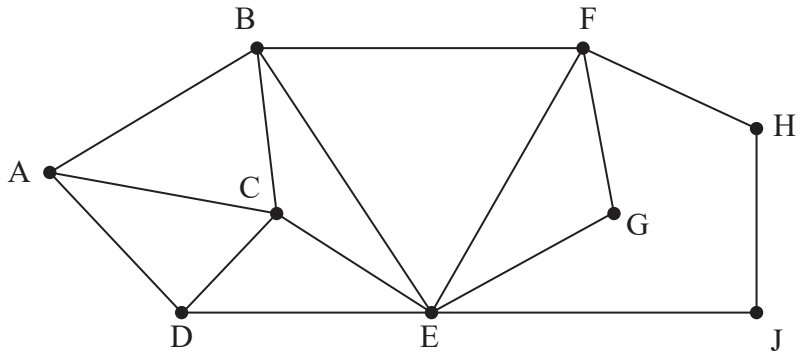


Figure 1

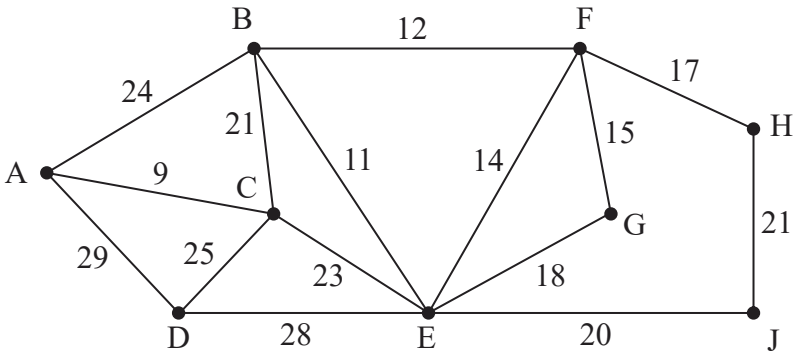


Figure 2

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Question 2 continued

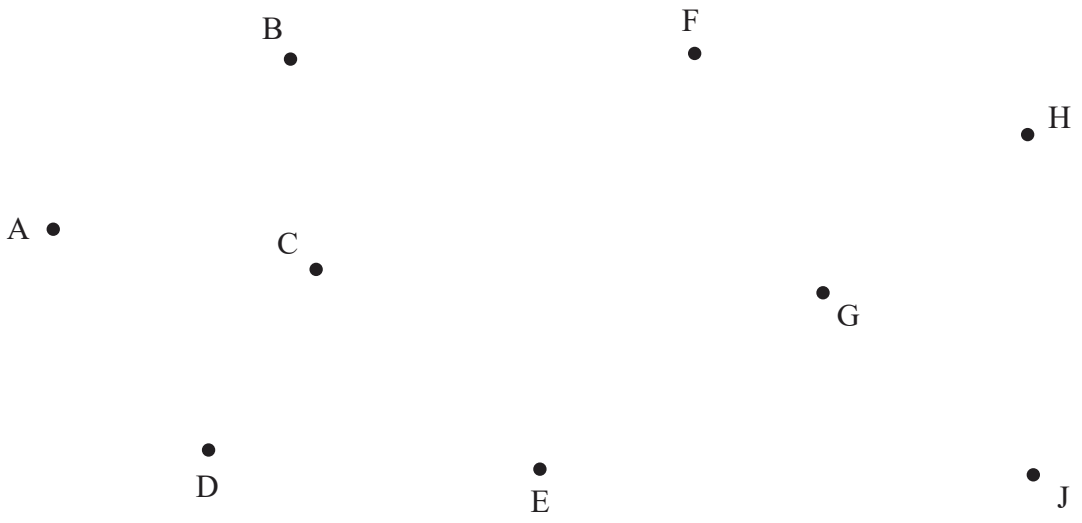


Diagram 1

Weight of the minimum spanning tree: _____

(Total 7 marks)

5.

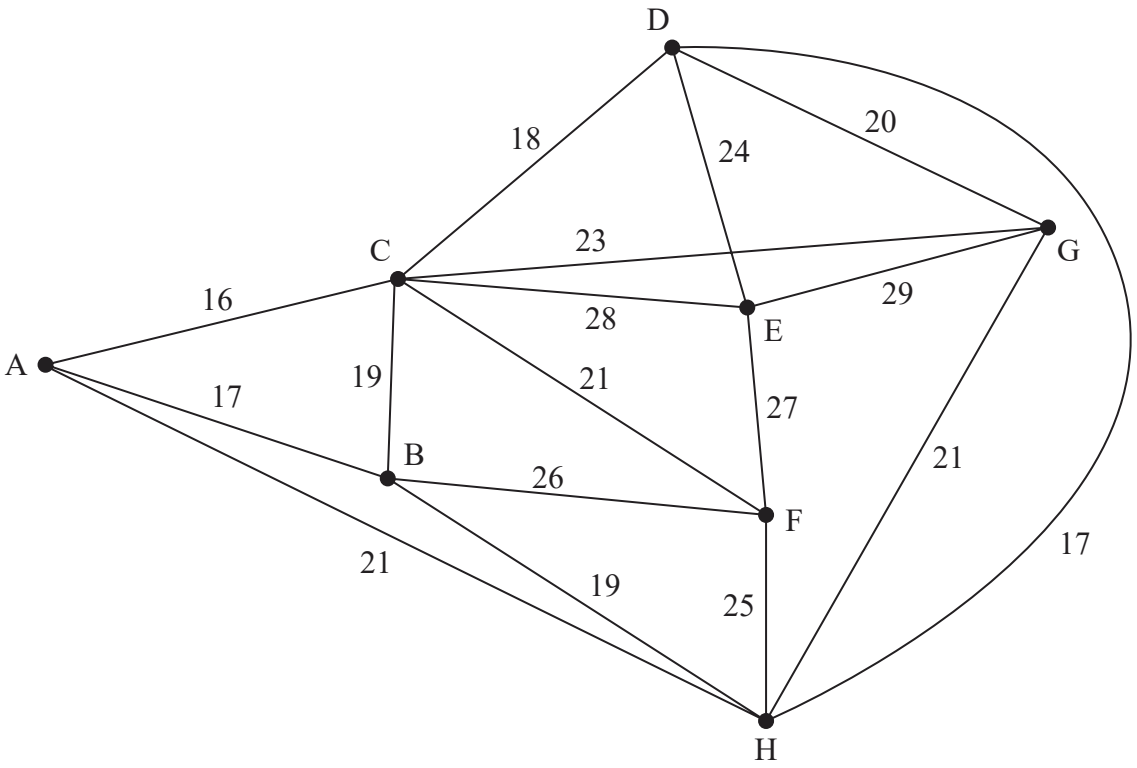


Figure 3

(a) Explain why it is impossible to draw a graph with eight vertices in which the vertex orders are 1, 2, 2, 3, 3, 4, 4 and 6

(1)

Figure 3 shows the network T. The numbers on the arcs represent the distances, in km, between the eight vertices, A, B, C, D, E, F, G and H.

(b) Determine whether or not A – C – D – E – C – B – F is an example of a path on T. You must justify your answer.

(2)

(c) Use Prim's algorithm, starting at A, to find the minimum spanning tree for T. You must clearly state the order in which you select the arcs of the tree.

(3)

(d) Draw the minimum spanning tree using the vertices given in Diagram 1 in the answer book.

(1)

The weight of arc CF is now increased to a value of x. The minimum spanning tree for T is unique and includes the same arcs as those found in (c).

(e) Write down the smallest interval that must contain x.

(2)

(Total for Question 5 is 9 marks)

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Question 5 continued

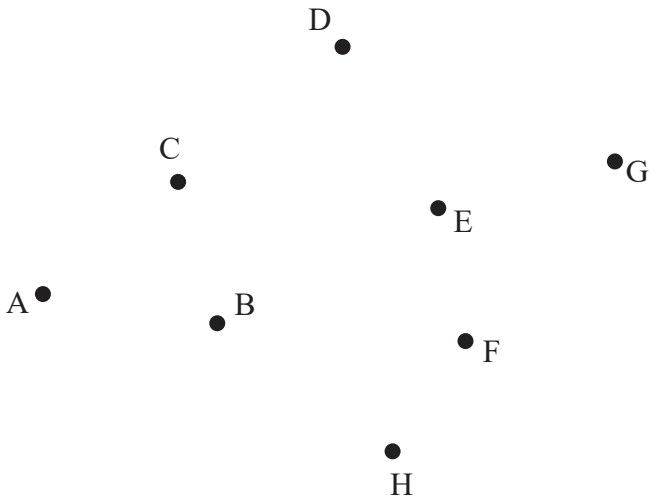


Diagram 1

(Total for Question 5 is 9 marks)

Write your answers in the D1 answer book for this paper.

1. The table below shows the distances, in metres, between six vertices, A, B, C, D, E and F, in a network.

	A	**B**	**C**	**D**	**E**	**F**
A	–	18	23	17	28	19
B	18	–	20	11	–	24
C	23	20	–	–	25	13
D	17	11	–	–	–	22
E	28	–	25	–	–	–
F	19	24	13	22	–	–

(a) Draw the weighted network using the vertices given in Diagram 1 in the answer book.

(2)

(b) Use Kruskal's algorithm to find a minimum spanning tree for the network. You should list the edges in the order that you consider them and state whether you are adding them to your minimum spanning tree.

(3)

(c) Draw the minimum spanning tree on Diagram 2 in the answer book and state its total weight.

(2)

(Total 7 marks)

Leave blank

1.

	A	B	C	D	E	F
A	–	18	23	17	28	19
B	18	–	20	11	–	24
C	23	20	–	–	25	13
D	17	11	–	–	–	22
E	28	–	25	–	–	–
F	19	24	13	22	–	–

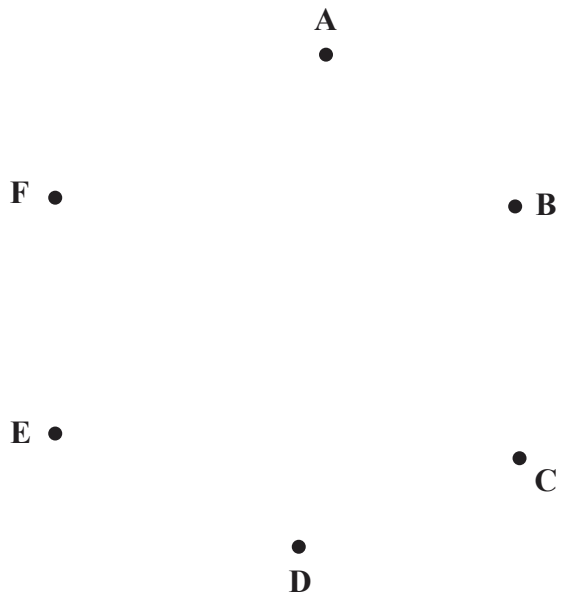


Diagram 1

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Question 1 continued

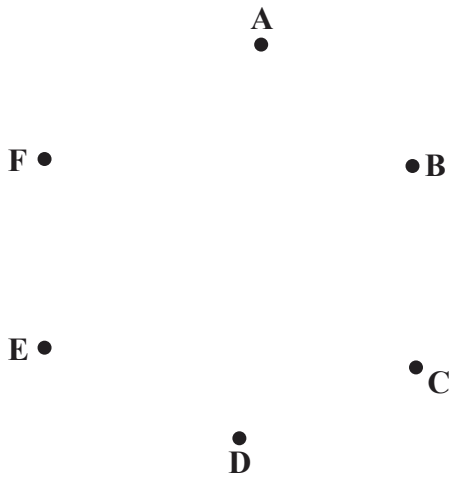


Diagram 2

Weight of minimum spanning tree _____

Q1

(Total 7 marks)

7.

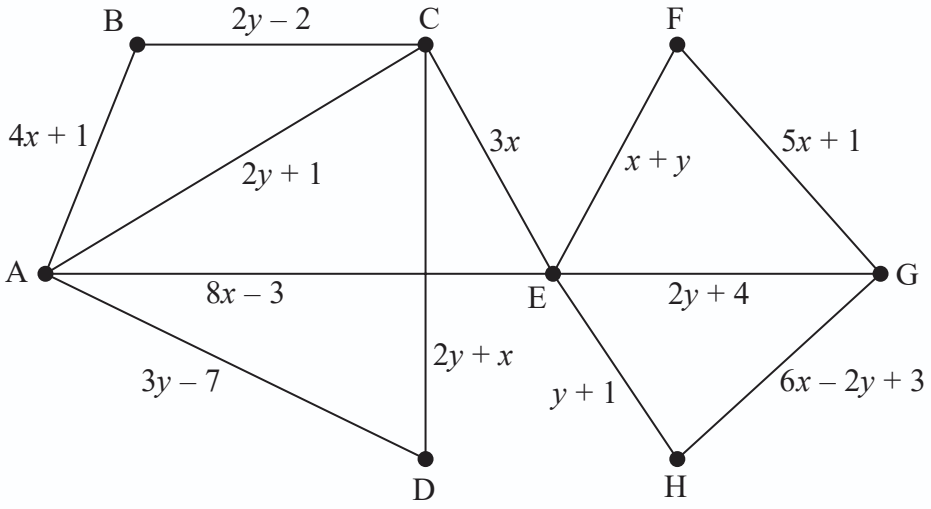


Figure 5

Figure 5 shows a weighted graph that contains 12 arcs and 8 vertices.

It is given that

- no two arcs have the same weight
- x and y are positive integers
- arc CD is not in the minimum spanning tree for the graph

(a) Explain why $y < x + 7$

(2)

It is also given that when Prim's algorithm, starting at A, is applied to the weighted graph, AB is the first arc selected.

(b) Show that $y > 2x$ and write down and simplify two further constraints on the values of x and y.

(3)

(c) Represent these four constraints on Diagram 1 in the answer book.

(4)

(d) Using Diagram 1 **only**, write down the possible pairs of values that x and y can take in the form (x, y).

(2)

The minimum spanning tree for the weighted graph in Figure 5 has total weight 73
Six of the seven arcs in the minimum spanning tree are AB, AD, BC, CE, EF and GH.

(e) Determine the value of x and the value of y. You must make your method and working clear.

(4)

(Total 15 marks)

TOTAL FOR PAPER: 75 MARKS

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Question 7 continued

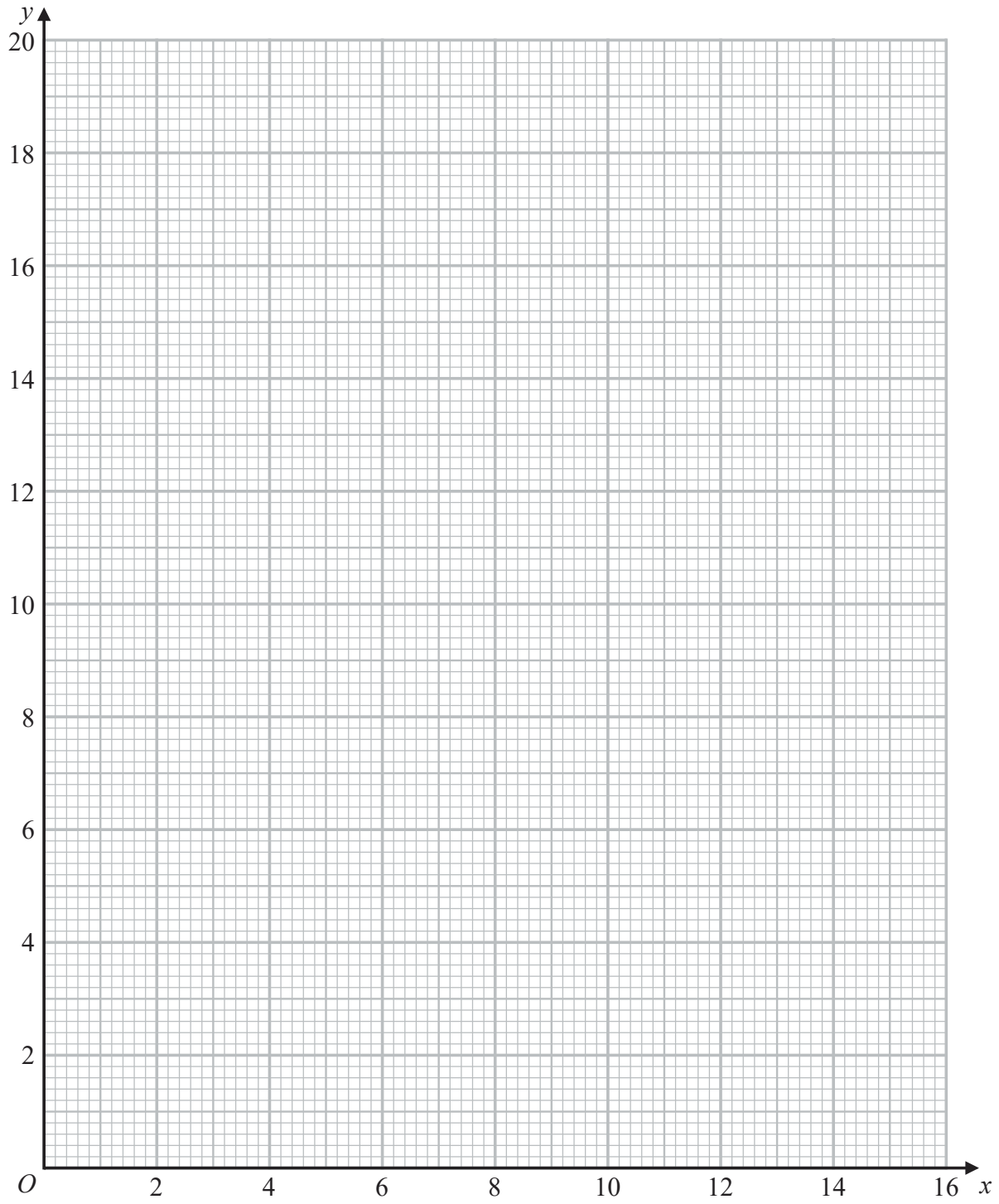


Diagram 1

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Question 7 continued

Q7

(Total 15 marks)

TOTAL FOR PAPER: 75 MARKS

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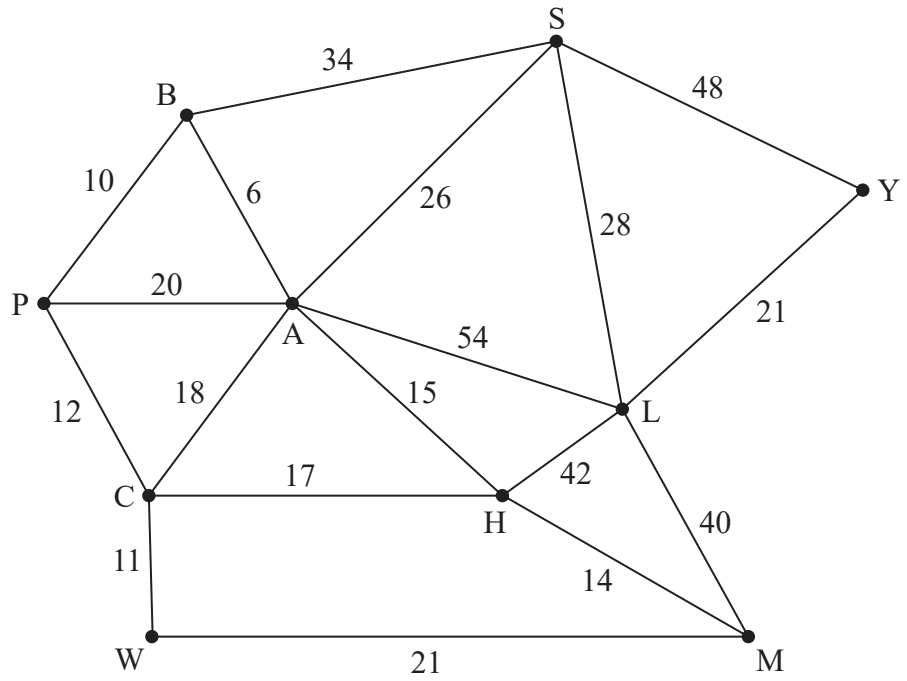


Figure 2

The network in Figure 2 shows the distances, in miles, between ten towns, A, B, C, H, L, M, P, S, W and Y.

(a) Use Kruskal's algorithm to find a minimum spanning tree for the network. You should list the arcs in the order in which you consider them. In each case, state whether you are adding the arc to your minimum spanning tree.

(3)

	A	B	C	H	L	M	P	S	W	Y
A	–	6	18	15	54	29	16	26	29	74
B	6	–	22	21	60	35	10	32	33	80
C	18	22	–	17	59	31	12	44	11	80
H	15	21	17	–	42	14	29	41	28	63
L	54	60	59	42	–	40	70	28	61	21
M	29	35	31	14	40	–	43	55	21	61
P	16	10	12	29	70	43	–	42	23	90
S	26	32	44	41	28	55	42	–	55	48
W	29	33	11	28	61	21	23	55	–	82
Y	74	80	80	63	21	61	90	48	82	–

The table shows the shortest distances, in miles, between the ten towns.

(b) Use Prim's algorithm on the table, starting at A, to find the minimum spanning tree for this network. You must clearly state the order in which you select the arcs of your tree.

(3)

(c) State the weight of the minimum spanning tree found in (b).

(1)

Sharon needs to visit all of the towns, starting and finishing in the same town, and wishes to minimise the total distance she travels.

(d) Use your answer to (c) to calculate an initial upper bound for the length of Sharon's route.

(1)

(e) Use the nearest neighbour algorithm on the table, starting at W, to find an upper bound for the length of Sharon's route. Write down the route which gives this upper bound.

(2)

Using the nearest neighbour algorithm, starting at Y, an upper bound of length 212 miles was found.

(f) State the best upper bound that can be obtained by using this information and your answers from (d) and (e). Give the reason for your answer.

(1)

(g) By deleting W and all of its arcs, find a lower bound for the length of Sharon's route.

(2)

Sharon decides to take the route found in (e).

(h) Interpret this route in terms of the actual towns visited.

(1)

(Total for Question 3 is 14 marks)

3.

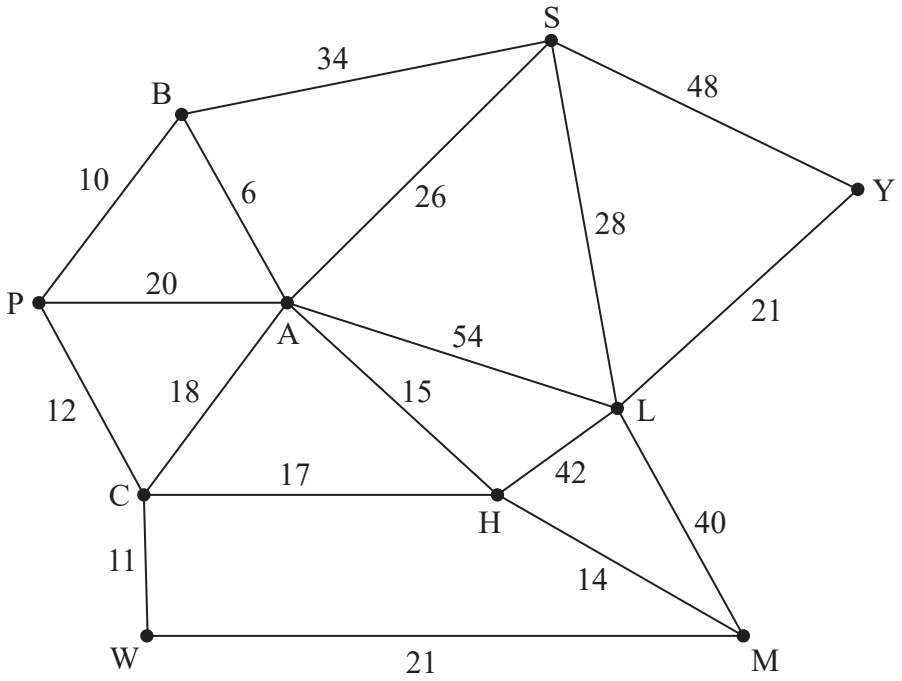


Figure 2

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Question 3 continued

	A	B	C	H	L	M	P	S	W	Y
A	–	6	18	15	54	29	16	26	29	74
B	6	–	22	21	60	35	10	32	33	80
C	18	22	–	17	59	31	12	44	11	80
H	15	21	17	–	42	14	29	41	28	63
L	54	60	59	42	–	40	70	28	61	21
M	29	35	31	14	40	–	43	55	21	61
P	16	10	12	29	70	43	–	42	23	90
S	26	32	44	41	28	55	42	–	55	48
W	29	33	11	28	61	21	23	55	–	82
Y	74	80	80	63	21	61	90	48	82	–

Question 3 continues on the next page.

Question 3 continued

	A	B	C	H	L	M	P	S	W	Y
A	–	6	18	15	54	29	16	26	29	74
B	6	–	22	21	60	35	10	32	33	80
C	18	22	–	17	59	31	12	44	11	80
H	15	21	17	–	42	14	29	41	28	63
L	54	60	59	42	–	40	70	28	61	21
M	29	35	31	14	40	–	43	55	21	61
P	16	10	12	29	70	43	–	42	23	90
S	26	32	44	41	28	55	42	–	55	48
W	29	33	11	28	61	21	23	55	–	82
Y	74	80	80	63	21	61	90	48	82	–

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Question 3 continued

	A	B	C	H	L	M	P	S	W	Y
A	–	6	18	15	54	29	16	26	29	74
B	6	–	22	21	60	35	10	32	33	80
C	18	22	–	17	59	31	12	44	11	80
H	15	21	17	–	42	14	29	41	28	63
L	54	60	59	42	–	40	70	28	61	21
M	29	35	31	14	40	–	43	55	21	61
P	16	10	12	29	70	43	–	42	23	90
S	26	32	44	41	28	55	42	–	55	48
W	29	33	11	28	61	21	23	55	–	82
Y	74	80	80	63	21	61	90	48	82	–

(Total for Question 3 is 14 marks)

4.

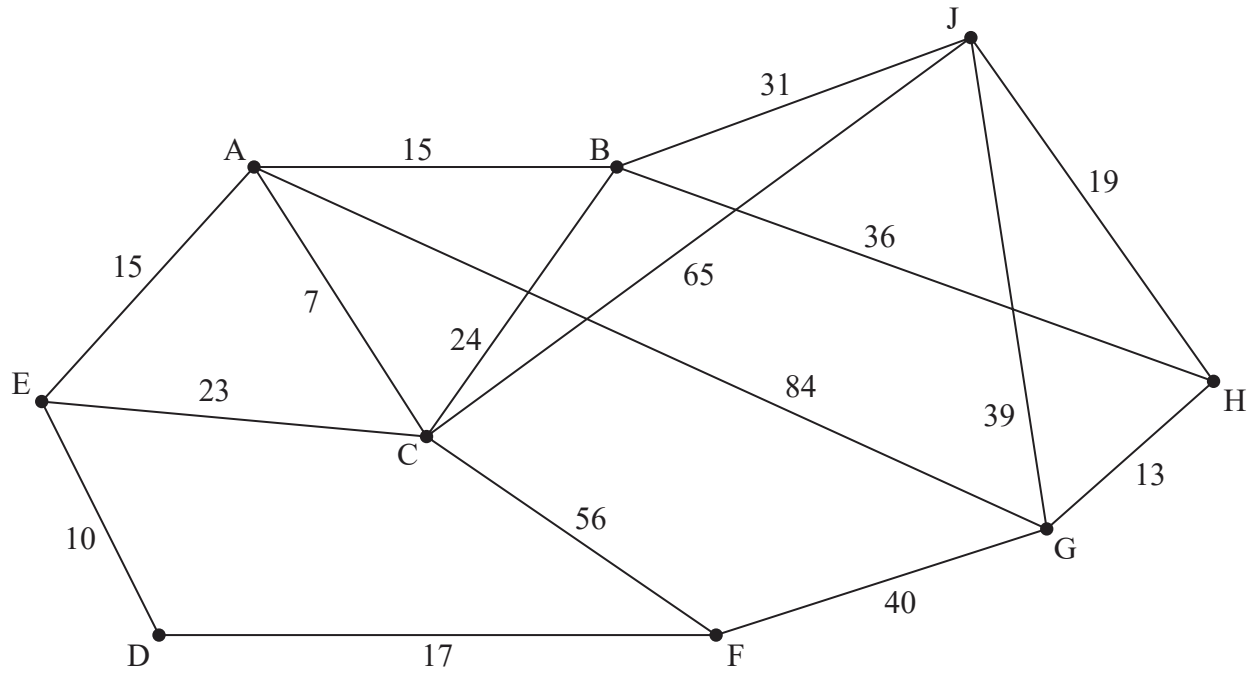


Figure 3

[*The total weight of the network is* 494]

Direct roads between nine factories, A, B, C, D, E, F, G, H and J, are represented in Figure 3.

The number on each arc represents the lengths, in kilometres, of the corresponding road.

The table below shows the shortest distances, in kilometres, between the nine factories.

	A	B	C	D	E	F	G	H	J
A	-	15	7	25	15	42	64	51	46
B	15	-	22	40	30	57	49	36	31
C	7	22	-	32	22	49	71	58	53
D	25	40	32	-	10	17	57	70	71
E	15	30	22	10	-	27	67	66	61
F	42	57	49	17	27	-	40	53	72
G	64	49	71	57	67	40	-	13	32
H	51	36	58	70	66	53	13	-	19
J	46	31	53	71	61	72	32	19	-

Table of shortest distances

(a) Starting at A, use Prim's algorithm to find a minimum spanning tree for the table of shortest distances. You must state the order in which you select the arcs of your tree.

(3)

(b) State the weight of the minimum spanning tree.

(1)

A route is needed that minimises the total distance to traverse each road at least once. The route must start at E and finish at F.

(c) Determine the length of this route. You must give a reason for your answer.

(2)

It is now decided to start the route at C and finish the route at A. The route must include every road at least once and must still minimise the total distance travelled.

(d) By considering the pairings of all relevant nodes, find the roads that need to be traversed twice.

(4)

Naoko needs to visit all nine factories, starting and finishing at the same factory, and wishes to minimise the total distance travelled.

(e) Starting at B, use the nearest neighbour algorithm on the table of shortest distances to find an upper bound for the length of Naoko's route. Write down the cycle, obtained from the table of shortest distances, which gives this upper bound.

(2)

(f) By deleting C and all of its arcs, use the values in the table of shortest distances to find a lower bound for the length of Naoko's route.

(2)

(Total for Question 4 is 14 marks)

4.

	A	B	C	D	E	F	G	H	J
A	-	15	7	25	15	42	64	51	46
B	15	-	22	40	30	57	49	36	31
C	7	22	-	32	22	49	71	58	53
D	25	40	32	-	10	17	57	70	71
E	15	30	22	10	-	27	67	66	61
F	42	57	49	17	27	-	40	53	72
G	64	49	71	57	67	40	-	13	32
H	51	36	58	70	66	53	13	-	19
J	46	31	53	71	61	72	32	19	-

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Question 4 continued

	A	B	C	D	E	F	G	H	J
A	-	15	7	25	15	42	64	51	46
B	15	-	22	40	30	57	49	36	31
C	7	22	-	32	22	49	71	58	53
D	25	40	32	-	10	17	57	70	71
E	15	30	22	10	-	27	67	66	61
F	42	57	49	17	27	-	40	53	72
G	64	49	71	57	67	40	-	13	32
H	51	36	58	70	66	53	13	-	19
J	46	31	53	71	61	72	32	19	-

Question 4 continued

	A	B	C	D	E	F	G	H	J
A	-	15	7	25	15	42	64	51	46
B	15	-	22	40	30	57	49	36	31
C	7	22	-	32	22	49	71	58	53
D	25	40	32	-	10	17	57	70	71
E	15	30	22	10	-	27	67	66	61
F	42	57	49	17	27	-	40	53	72
G	64	49	71	57	67	40	-	13	32
H	51	36	58	70	66	53	13	-	19
J	46	31	53	71	61	72	32	19	-

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Question 4 continued

	A	B	C	D	E	F	G	H	J
A	-	15	7	25	15	42	64	51	46
B	15	-	22	40	30	57	49	36	31
C	7	22	-	32	22	49	71	58	53
D	25	40	32	-	10	17	57	70	71
E	15	30	22	10	-	27	67	66	61
F	42	57	49	17	27	-	40	53	72
G	64	49	71	57	67	40	-	13	32
H	51	36	58	70	66	53	13	-	19
J	46	31	53	71	61	72	32	19	-

(Total for Question 4 is 14 marks)



Write your answers in the D1 answer book for this paper.

1. (a) Explain what is meant by the term 'path'.

(2)

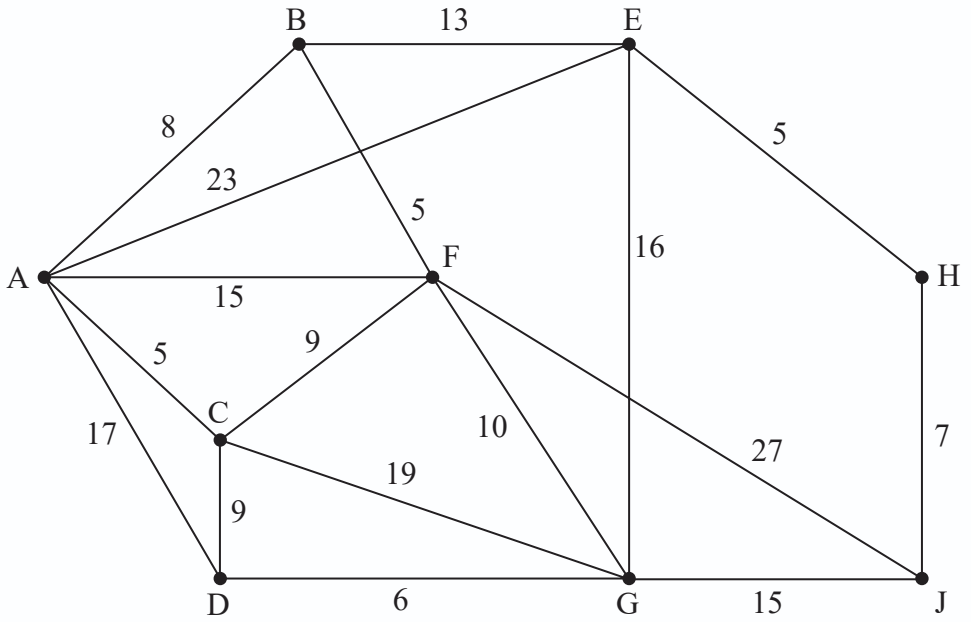


Figure 1

Figure 1 represents a network of roads. The number on each arc represents the length, in km, of the corresponding road. Piatrice wishes to travel from A to J.

(b) Use Dijkstra's algorithm to find the shortest path Piatrice could take from A to J. State your path and its length.

(6)

Piatrice needs to return from J to A via G.

(c) Find the shortest path Piatrice could take from J to A via G and state its length.

(2)

(Total 10 marks)

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1.

(a) _____

_____

_____

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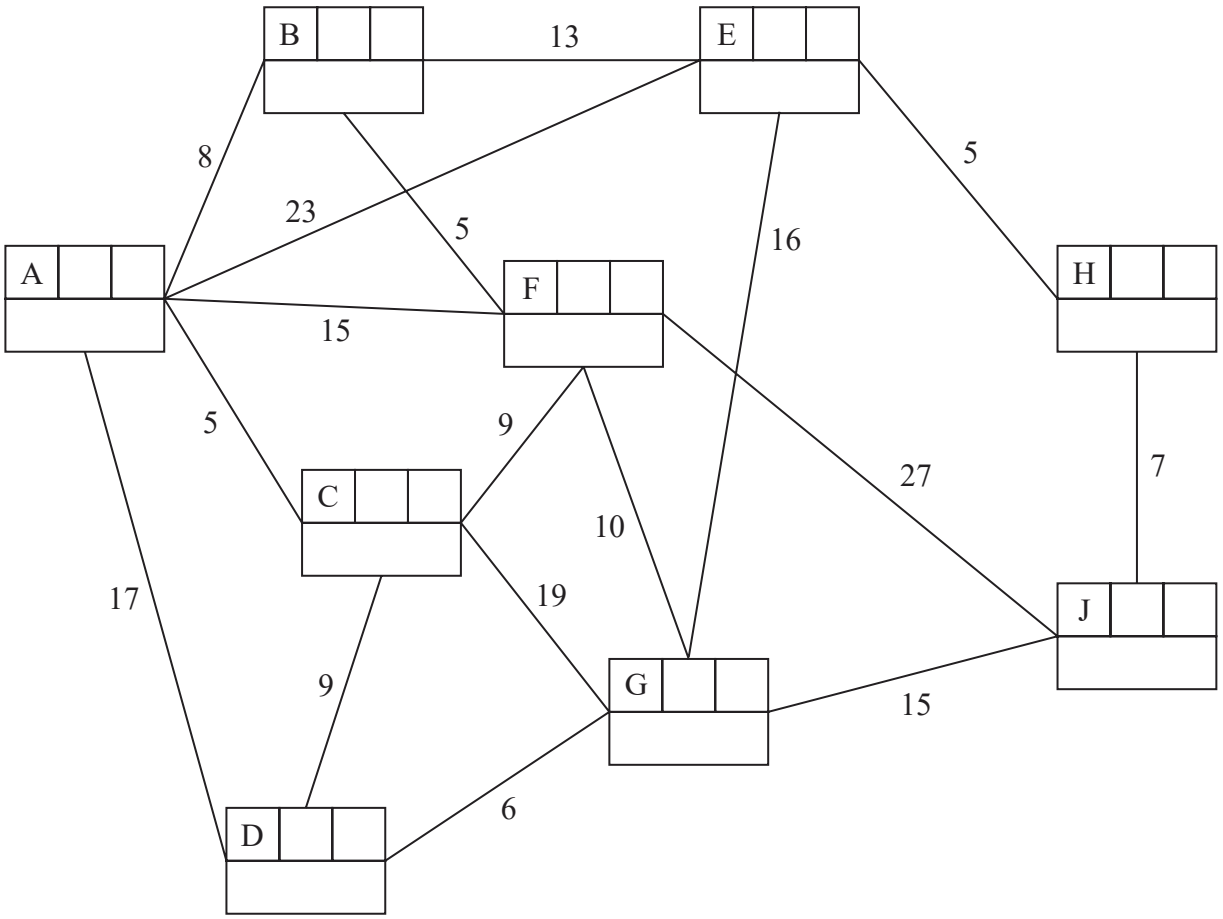
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Key:

Vertex	Order of labelling	Final value
Working values		



Shortest path from A to J: _____

Length of shortest path from A to J: _____

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Question 1 continued

(Total 10 marks)

Q1